



Incorporating parental values in complex paediatric and perinatal decisions

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Incorporating parental values in complex medical decisions for young children is important but challenging. In this Review, we explore what it means to incorporate parental values in complex paediatric and perinatal decisions. We provide a narrative overview of the paediatric, ethics, and medical decision-making literature, focusing on value-based and ethically complex decisions for children who are too young to express their own preferences. We explain key concepts and definitions, discuss paediatric-specific features, reflect on challenges in learning and expressing values for both parents and health-care providers, and provide recommendations for clinical practice. Decisional values are informed by global and external values and could relate to the child, the parents, and the whole family. These values should inform preferences and assure value-congruent choices. Additionally, parents might hold various meta values on the process of decision making itself. Complex decisions for young children are emotionally taxing, ethically difficult, and often surrounded by uncertainty. These contextual factors make it more likely that values and preferences are initially absent or unstable and need to be constructed or stabilised. Health-care professionals and parents should work together to construct and clarify values and incorporate them into personalised decisions for the child. An open communication style, with unbiased and tailored information in a supportive environment, is helpful. Dedicated training in communication and shared decision making could help to improve the incorporation of parental values in complex decisions for young children.

Introduction

Making complex medical decisions for fetuses, neonates, and infants is ethically and practically challenging. Incorporating the values of patients (or their parents) in both adult and paediatric medicine is essential^{1–10} as part of shared decision making and personalised health care.^{2,5,11–13} However, in practice, values are not always well elicited or incorporated into decisions.^{3,14–21}

Decision making in paediatrics is different from adult medicine.^{3,4,22,23} Although adults can decide for themselves, fetuses, neonates, and infants cannot; therefore, parents or guardians are surrogate decision makers. Having parents be part of decision making is usually beneficial; Emanuel and Emanuel have argued that a deliberative physician–patient relationship would best promote the autonomy and wellbeing of patients.²⁴ Some empirical support is available for this contention. One study described parents who shared in neonatal end-of-life decisions experienced less grief after a few years compared with parents who did not.²⁵ Other parents expressed that although participating in decision making was emotional and burdensome, it enabled them to prepare for or prevent a feared situation, and to feel more empowered.^{26,27}

In this Review, we explore what it means to incorporate parental values into complex paediatric and perinatal decisions. For readability, we use the term parents, but we mean to refer to people who have responsibility for a child, such as parents or guardians. We provide a narrative overview incorporating literature from paediatrics, ethics, and medical decision making. Our focus is on value-based decisions in which no single best option exists for children who are too young to express their own preferences on ethically complex issues. Such decisions are also sometimes referred to as falling into the grey zone, the zone of parental

discretion, or as preference-sensitive decisions; by definition, these decisions are ones that parents are permitted to make.^{1,2,6,9,28–32} Such decisions might feature prognostic uncertainty, insufficient evidence on harms and benefits, or a very personal benefit–harm ratio.^{2,12,32} Examples can be end-of-life decisions or advanced care planning in life-limiting diseases.^{6,33,34} Identifying whether decisions fall within the grey zone of parental discretion can be challenging; in this Review, we focus on cases that clearly fit into this category.^{6,7,23,35–38}

A single paediatric definition of shared decision making is missing, and alternative terminology has been proposed.^{3,5,6,12,21,27,39} However, a commonality between

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Key messages

- Incorporating parental values in complex medical decisions for young children is crucial and demanding.
- Decisional values are shaped by global and external values, encompassing aspects related to the child, the parents, and the entire family. These values should guide preferences and assure value-congruent choices.
- Parents might also hold various meta values regarding the decision-making process itself.
- Given the emotionally taxing, ethically challenging, and often uncertain nature of complex decisions for young children, values and preferences could initially be absent or unstable, requiring construction or stabilisation.
- Collaborative efforts between health-care professionals and parents are essential to construct and elucidate values, integrating them into personalised decisions. An open communication style, and delivering unbiased and tailored information in a supportive environment, proves beneficial.

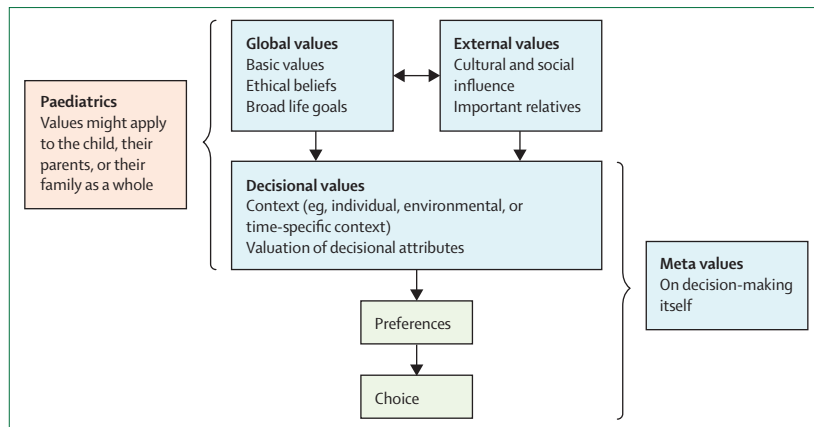


Figure: Overview of concepts and relations in value-based decision making for fetuses, infants, and young children

definitions is the importance of clarifying parental values and incorporating these into value-congruent decisions—choices that align with the values of the people most affected by the decision.^{6,13,40}

Key concepts

The concept of values generally refers to what is good and worthy and is often used in medical decision making, yet values are often left undefined. In psychology, personal values are “broad desirable goals that motivate people’s action and serve as guiding principles in their lives across situations. They affect people’s preferences and behaviour over time and across situations”.⁴¹ In research on patient decision aids, definitions for values can be similarly wide-ranging, such as “an umbrella term referring to what matters to an individual relevant to a health decision”.⁴⁰ However, the concept of values is multifaceted (figure). Key components include global or basic values (ie, underlying values, ethical beliefs, or life goals, which exist beyond specific decisions but influence decision-specific values), external values (ie, values beyond the individual that a person wants to take into consideration, such as societal values, cultural values, and values of important relatives and friends), and decisional values (ie, values that are at least partly specific for a decision). Decisional values include decision-specific valuation of attributes and incorporate specific (and potentially transient) contextual factors (eg, specific to the individual, environment, or time).^{42,43} Separate from these values relating to the decision itself are meta values relating to the decision-making process, such as how an individual wants to make decisions and what their responsibility or role should be. Finally, values exist that are specific to paediatrics and might apply to the child, their parents, other family members (eg, siblings), or to the whole family.

Several other terms are often used interchangeably,^{44–46} such as goals, preferences, perspectives, arguments, views, and attitudes. One important distinction is the

idea that values are broader and more fundamental, whereas preferences are specific and directly related to the health-care decision. Therefore, a preference describes the extent to which an option is desirable, either abstractly or in comparison to an alternative.⁴⁰ This preference is influenced by decisional values, which are in turn influenced by global and external values (panel 1).

Unlike the field of psychology, in which four higher-order values are quite well defined (ie, conservation, self-enhancement, openness to change, and self-transcendence), in medicine and particularly paediatrics, values are not well characterised.⁴¹ In a report from a children’s hospital ethics committee, the three most important factors influencing parental decisions were the child’s quality of life, the chance of getting better, and their degree of discomfort.^{48,49} A paper on advanced care planning in children with rare diseases reported five domains expressed by parents: getting out and moving freely, feeling included and engaged, managing symptoms and disease burden, optimising coordinated care among many team members, and planning for the future.⁵⁰ Some papers identify the influential values of being a good parent and protecting the child from harm.^{5,22,27,51–53}

Paediatric-specific features of decision making

Parents or guardians are default surrogate decision makers for young children.^{3,4,12,22,53} In most countries, they also have legal decision-making responsibilities that differ from surrogates in adult decision making.^{5,19,22} Paediatric decision making is more complicated than adult decision making, partly because there is usually more than one parent with decision-making responsibilities.²³ Each parent might vary in their views, values, and ability to process information and participate in the decision-making process.¹⁰ Choosing for oneself versus for others influences treatment preferences.³ Various perspectives could have a role in surrogate decision making, such as egocentric, projected, benevolent, or substituted ones.⁵⁴ For very young children, substituted decision making (ie, knowing what a person would want in specific circumstances) is not applicable because they are too young to express preferences.^{5,22} Although parents are used to making decisions for their children, shared medical decision making can cause stress, anxiety, and an emotional burden for surrogates.^{19,55}

Some unique features of decisions for young infants include potential perceived social pressures (eg, a view that children should not die).^{5,22} Decisions can carry a heavy ethical weight because of the potential for long-term risks and benefits, combined with a relatively high amount of uncertainty about prognosis.^{8,19,32,56,57} In the perinatal period the patient might not have even been born, and their parents might not yet feel like parents. Decisions might need to be made in a sensitive period of parent–child bonding, when a firm place for the child

within the family or social structure might not have been established.⁵⁸ In addition, parental autonomy is more restricted than adults' autonomy to make decisions for themselves, as decisions that would cross the threshold of harm and risk substantial harm for their child are not (or should not be) permitted.⁵⁹ Notwithstanding the clearly serious effect of such potential harms, they fall outside the scope of this Review.

Challenges in learning and expressing values for parents and health-care professionals

An overview of challenges and potential strategies in knowing and clarifying values can be found in the table. Global values develop throughout childhood and adolescence, remain relatively consistent during adulthood, and reflect "a lens through which patients view all decisions".^{41,42} These values could be influenced by factors such as personality or religion.⁴² Parental values are relevant because of the potential similarity of values between parents and their children, both directly (through transmission) and indirectly (through shared environment, culture, and socioeconomic status).⁴¹ However, existing evidence suggests that health-care professionals often concentrate more on biomedical decisional values and preferences than on global psychological, spiritual, and social values.^{20,26,60}

Depending on the context, parents' values could also be influenced by family, school staff, other parents, and community members.^{3,42} Distinguishing between unwanted external pressure and situations in which parents involve others in medical decision making "based on ongoing relationships, reciprocal concern, respect of others' advice, and mutual interests without implying coercion or undue influence"⁴² can be challenging.

Value construction and context

Parents might have pre-existing decisional values and preferences (eg, from facing similar decisions before). However, for complex medical decisions, values probably need to be uncovered or even constructed at the time of decision making.³⁷ This need raises a further problem: knowing one's values and preferences in ethical decisions is not easy, especially when affected by high-stake consequences, conflicting priorities, high amounts of uncertainty and emotion, little familiarity with the decision, and time pressure.^{23,61–63}

A range of contextual factors influence the expression of values and preferences, including the adequacy of information, credibility of information source, clarity of communication, choice architecture (ie, the way choices are presented or framed), emotions, perceived social norms, and comfort (or discomfort) with the medical environment.^{1,3,23,27,37,55,57,61,62,64,65}

Three important factors influencing the expression of values and preferences are knowledge, bias, and emotions, especially in the relationship between a health-care professional and parent.^{1,58,62,65–67} Health-care

Panel 1: Concepts and definitions

- Value: an umbrella term referring to what matters to an individual relevant to a health decision. Values might be directly relevant to decisions (eg, "beliefs, feelings, or perceptions regarding attributes of a treatment option"⁴⁰) or indirectly relevant (eg, goals, world views, or family, religious, or cultural values). Values might be represented qualitatively or, in some cases, quantitatively. This definition is deliberately broad.⁴⁰
- Value clarification: the process of establishing what matters to an individual relevant to a given health decision. This definition emphasises that what matters to an individual might be broader than attribute-specific values. What matters could also include preferences, concerns (eg, concerns about changes in health status), and issues to do with the context of a person's life within which they would need to implement a decision (eg, fitting a treatment plan into one's work schedule).⁴⁰
- Value clarification methods: strategies that are intended to help patients evaluate the desirability of options or attributes of options within a specific decision context to identify which option they prefer. These methods can be implicit or explicit:⁴⁰
- Implicit: strategies for facilitating value clarification that do not require people to interact with anything or anyone—for example, describing "options in enough detail that clients can imagine what it is like to experience the physical, emotional, and social effects",⁴⁰ or simply encouraging people to think about what matters to them.
- Explicit: strategies for facilitating value clarification that require people to interact with something or someone (eg, completing a worksheet, using an interactive website, having a semistructured conversation with another person with the explicit purpose of clarifying values, or engaging in another structured exercise).⁴⁰
- Preference: the extent to which a decision option or health state is desirable or acceptable, either in the abstract context or compared with other options or health states. Preferences might be represented qualitatively or, more commonly, quantitatively.⁴⁰
- Choice: the selected option, either intended or actual, for screening or treatment decisions.⁴⁴
- Value congruency: a calculation of the match between the chosen option or choice (ie, the dependent variable) and the patient's values (ie, the independent variable) or, in other words, ensuring that the decision aligns with what matters to the people most affected by the decision.^{44,47}

workers should provide all relevant information about a child's condition and treatment options, and effort should be made to help parents understand this.^{23,37,57}

Low levels of health literacy or numeracy can make it difficult for parents to express preferences in response to graphs and probabilities.^{2,4,23,64,68} Health-care professionals do not always assess parents' level of understanding.⁶⁴ Irrelevant information might be given, and current or future factors worth considering might be selectively omitted.^{1,37,62,67} Health-care professionals might introduce bias, consciously or unconsciously, and bring their own values into the conversations. Influences of choice architecture include framing options by emphasising, for example, survival rates (positive framing) or mortality (negative framing) or using the order by which options are presented to suggest relative differences (eg, offering a default treatment before mentioning alternative options). Emotions can make understanding and interpreting information difficult and could alter parents' perceptions.^{1,57,62} Feelings of guilt or failure,³ or feeling

Potential strategies for health-care professionals and the health-care system	
Values of parents	
Using parental values as a surrogate for the child who is too young to have or express their own values	Health-care professionals should allow some parental interest (ie, acknowledging that the interests of parents and their children are linked), as values exist similarly between parents and children
Value construction (parents)	
Features that make values and preferences less likely to pre-exist, and therefore need to be constructed, include high-stake consequences, conflicting values, high amounts of uncertainty, little familiarity with the decision, emotionally charged environments, and time pressure	Health-care professionals should help parents construct decisional values (eg, by asking open-ended questions on what parents hope and fear, encouraging them to share intuitions, reflecting on values, and gently challenging and translating these values into preferences); facilitate a two-way information exchange (from health-care professional to parents on diagnosis, prognosis, and trajectory of their child's condition, and from parents to health-care professional on their situation); and empower parents to participate in the decision-making process
Context	
Contextual factors that influence the expression of decisional values and preferences include adequacy of information and knowledge, credibility of information source, clarity of language, attention to different attributes of a decision, caregiver responses, choice architecture, emotions, social norms (perceived or actual), and comfort (or discomfort) with the medical environment	Health-care professionals should be aware of their own biases; provide information that is tailored and sensitive to choice architecture; use tools for information dissemination if appropriate; use expertise, literature, and experience to be informed about common attributes that could be of importance; be aware of emotions (address them or use a built-in delay or distraction); create a safe and quiet place and be present; allow parents to postpone a decision or make a provisional decision; guarantee a follow-up
Conflicting values and preferences	
Conflicting values within one person, between parents, between the parent and child, public and child, or process and outcome	Health-care professionals should address different values between stakeholders
Meta values	
Role preferences in decision making (ie, parents, health-care professional, or both) can be contextual and are not always accurately predicted by the health-care professional	Health-care professionals should communicate about the decision-making process and be sensitive to cues; providing guidance upon request can be appropriate
Preferences regarding the decision-making process can vary (eg, they can be more or less rational or intuitive)	Health-care professionals should be aware of pitfalls of rational deliberation (eg, intensifying emotions, selective extraction, or simplifying information) and intuition (eg, risks of bias and heuristics, early uninformed decisions, insufficient reasoning); and gently challenge preferences
Interaction between health-care professionals and parents on values	
Patients rarely discuss their values directly	The health-care professional should adopt an active role in clarifying values (choosing to work from a decision and its options to values, or from values towards specific preferences and choices); a decision aid including value clarification methods could be useful
Several barriers influence the discussion about values between health-care professional and parent, including insufficient (perceived) parental readiness, health-care professionals assuming they know the parents' values and preferences, little time in discussions, insufficient skills, poor relationships between the health-care professional and parents, not having a regular point of contact, ineffective communication, strong emotion, health-care professionals being negatively biased towards value clarification, not having a good way to evaluate value congruence for health-care professionals, insufficient support systems (eg, guidelines, tools, and organisation)	Health-care professionals should ask parents for permission to discuss decisions; ask for and provide feedback on conversations (eg, peer-to-peer coaching and mirror interviews); start early with conversations and repeat conversations if necessary; and appoint a contact person; in turn, health-care systems should educate health-care professionals on value clarification skills, ensure a supportive health-care system (guidelines should be sensitive to value-based decisions, describe trade-offs, and allow time for encounters), and develop tools such as patient decision aids with expertise from both health-care professionals and parents

Table: Overcoming challenges in incorporating parental values in complex paediatric and perinatal decisions

anxious, overwhelmed, in denial, or defensive, could hinder their ability to express values and preferences.²³

Conflicting values and preferences

Conflicts can arise between values, such as the tension between prolonging a child's life and maximising their comfort. Such conflicts can occur within one parent or between parents, and translation into specific preferences can be difficult.^{27,37,42,68} Furthermore, specific preferences can be in conflict with underlying values, such as when a preferred intervention is incompatible with the valued outcome or when preferences are open to interpretation (eg, a parent asking for everything to be done to help their child).^{64,69} In paediatrics, potential conflicts exist between values relating to the wider family versus the child.^{54,70} Moreover, parental values are inextricably linked or entangled with those of the child.^{32,70} A family's individual values might also be conflicting with those of their community or society.^{7,38}

Meta values

Most parents prefer to have a role in decision making, but the size of this role might depend on personality and context.^{3,4,12,19,33,57,71–80} In adults, a mismatch between actual and preferred roles in decision making is associated with increased anxiety and reduced satisfaction.⁵⁷ Parents generally prefer more responsibility when making decisions that are high risk, when they have experience with the decision, when the decision is thought to be part of the normal parental role, when wider goals are involved (eg, palliative care), and when decisions have the potential to harm their child.^{3,6,33,56,80}

In one study, having other children was associated with parents' preference for making the final decision, while another study found that previous paediatric intensive care admissions and having private insurance were associated with a preference for more medical influence.^{33,80} However, Mack and colleagues found no predicting parental characteristics.⁷⁹ Parents have also expressed

a desire for increased responsibility from health-care professionals in decisions that are urgent, require strong medical expertise, and could benefit an infant.^{33,56,80} The preference for the decision-making responsibility is also culturally influenced; for example, some cultures do not encourage shared decision making and prefer health-care providers to make decisions.^{19,81} Health-care professionals are often unable to predict the role preference of patients or their parents in the decision-making process, and false assumptions are common.^{3,27,67,73,79,82,83} Although role preferences regarding parents' final decision-making responsibilities vary, and some parents might not (yet) be ready for a conversation about values, they probably do not want their values to be ignored.

Parents also differ in how they wish to make complex decisions. Some prefer a more rational or deliberative process that is effortful, conscious, and analytical.^{63,65} Others prefer a less rational and more intuitive process.^{63,65} Explicit deliberation helps parents verbalise and articulate their preferences and reasons, make congruent decisions, and feel better about how the decision was made, allowing explicit consideration of the likelihood of outcomes and their values. However, negative emotions can intensify during this deliberation and parents might find reasons for preferences that are inaccurate because they are not congruent to their values, and they might restrict the decision-making process to those decision attributes that are easiest to verbalise while overlooking others.^{1,63,65,69} A more intuitive deliberation process might help in implicit integration of large amounts of information, as intuitive feelings can be surprisingly accurate. The risks of such intuition are the influence of bias (although this could also apply to rational processes), insufficient reasoning to help convince others and oneself that the decision is good, and, in early stages of decision making (eg, information gathering), uninformed decisions.⁶³

Interaction between parents and health-care professionals

Value clarification is often suboptimal in practice for both adults and children.^{43,60,84} Three studies analysing audiotaped conversations with surrogates on adult, paediatric, and neonatal intensive care units showed that a shared approach and value discussions were rare (although more common on neonatal and paediatric than on adult intensive care units) and that family responses were mostly passive.^{16,19,21}

Several problems have been identified for value clarification. Patients, including parents, rarely discuss their values directly.^{26,85} Barriers are numerous: insufficient perceived parental readiness, health-care professionals' assumptions about parental values and preferences, short amounts of time in clinical encounters, ineffective communication skills and related training, poor relationships between health-care professional and parents, not having a regular point of contact, bad communication,

heightened emotions (on both sides), health-care professionals' negative bias towards value clarification, ineffective methods for health-care professionals to evaluate value congruency, and an unsupportive health-care system (in terms of guidelines and tools).^{6,8,23,26,34,55,67,68,83,85,86}

Towards clinical practice

Parental values are not incorporated into paediatric decision making in one single correct way. Such incorporation will vary depending on the type of choice to be made, the information parents want to have, the role parents wish to have in the choice, and how rational parents can or want to be in the decision-making process.

Information exchange

Parents should be encouraged to become informed about their child's situation.^{63,64,68} They should understand the diagnosis, prognosis, and trajectory of an illness, although this understanding requires sensitivity to their needs.^{68,87} Uncertainty should be addressed.^{13,88} Parents, in turn, should provide information on their personal situation. Outcome parameters known to be relevant for both health-care professionals and parents should be part of the decision-making process.^{89,90} If available, a list of important themes about a given condition could serve as a useful starting point for parents who are inexperienced and simply want to gain understanding but might not know what information to ask for.⁹¹

Starting points for value clarification

The approach to value clarification depends on the context and preferences of those involved. If a well defined decision must be made, health-care professionals and parents might work towards the decision by discussing choices before values, or vice versa. In medically high-risk situations where a choice is yet to be made, health-care professionals generally start with a more general discussion of values.

Starting from a specific decision means acknowledging the decision at stake, including the options and their pros and cons. Ideally, parents and health-care professionals should work together from uninformed to informed preferences for a specific decision, reflecting a deeper understanding of the choices available.⁶⁸ Health-care professionals should allow both intuitive and rational deliberation, explore specific health beliefs, and help to optimise internal mental models.^{37,64} Iteration is needed for this dynamic, non-linear process with values and preferences unfolding that are influenced by emotions.^{37,64} Parental understanding of both the medical facts and their own values should be checked. For all decisions—especially the most intuitive ones—it could be crucial to allow time and delay before proceeding with an initial option.⁶⁴ An advantage of this approach is that parents are more likely to appreciate that there is a decision to be made and that their input is important.^{2,72} The approach also allows for paediatricians to establish whether or not

the decision is preference-sensitive before deliberation, and to involve parents only when an actual decision needs to be made, which avoids the problem of “speaking of decisions if there is nothing to decide yet or nothing to decide anymore”.^{6,12,92} In addition, parental autonomy in making decisions is more easily preserved when discussing choices before values.

However, starting with an exploration of values is common in paediatric advanced care planning or palliative care to better tailor care to individual families and to be better prepared when decisions arise.³⁴ An advantage of this approach is that choices are sometimes defined (or not) by specific values, especially when these values are less uncommon than others. Alternative choices could be overlooked if conversations start from predefined decisions.^{6,38,42} Even with a well defined decision at stake, working from value to choice is readily applicable. Parents should think about their own values and be given some understanding of their child’s medical condition, and health-care professionals should help translate these values into a specific preference.⁶⁸ This approach recognises that both the parental and health care-professional’s inputs are important, that patients can meaningfully engage without requiring a 100% understanding of the (complex) situation, and that parental autonomy is preserved in deciding “how life preferably should unfold”.⁶⁸ Furthermore, starting with parental values could be more appropriate for parents who prefer not to feel responsible for the final decision, and for parents with incomplete preferences.^{34,65,93}

Communication about values

Before discussing values, it can be beneficial to check whether parents are ready for a conversation and to empower them to participate, just as to address their meta values on decision making (suggestions by experts and from our own clinical experience are listed in panel 2).^{1,8,55,56,64,69,83} Some parents directly indicate their preferred role, but others do not or might do so indirectly (eg, by asking the doctor what they recommend).^{12,27,56,69} Providing guidance upon request is not paternalistic; if this guidance is sensitive to role preferences, it can be culturally sensitive and helpful for parents who feel burdened by the decision.^{69,76,81,93} Meta values might also be conflicting, such as when the wish for autonomy in decision making seems to result in a non-value-congruent decision.⁸⁶ Furthermore, parents could differ in how rational or intuitive they want the decision-making process to be. Therefore, health-care professionals should aim for acceptable (instead of perfectly rational) decisions.⁶⁵

To prevent strong emotions from impeding a conversation about values, addressing or sharing emotions first is often advised, as is building in a delay or distraction before deliberation.^{1,63,64,69} Assessing for emotional distress is recommended as psychological support could be helpful.⁵⁷ Health-care professionals

should actively listen; watch for cues in parents’ body language, faces, or voices; and remain silent at appropriate times.^{1,56,64,79,83} In addition, clinicians should actively work not only to elicit parents’ values, but also to help parents construct them. Open-ended questions should reflect on what parents are hoping for, on what they fear, and on what makes life meaningful to them and their children. Health-care professionals can ask parents about their specific concerns and hopes related to each option (if applicable), what is most important to them, and whether they need further information.^{1,56,58,68,83} Parents should be encouraged to also share their feelings towards options, even if those feelings are difficult to explain.^{63,64} Parents can be helped with prioritising conflicting values by being asked what good parenting means to them.⁸³

Health-care professionals should reflect on and check parental values by using the parents’ own words to confirm their understanding.^{1,13,56,64,68,83} Clear differences between stakeholders (eg, between parents or between parent and health-care professional) should be addressed.⁶⁴ Furthermore, health-care professionals might want to gently challenge a parental preference to establish whether it is stable and informed. The clinician can do this by asking whether the preference is really what the parents want after learning about the values that are important to them, encouraging them to postpone judgement to form an overall impression first. The clinician can also suggest that the parents’ preferences could change after learning more about risks and benefits of a decision, allowing doubt about a decision, and can help uncouple short-term positive or negative emotions from long-term effects of a decision.^{63,64,69}

Health-care professionals and parents should work together to translate values into a value-congruent choice. Again, using parents’ own words is useful, as is finding commonalities between beliefs and offering non-judgemental support.^{1,56,60,69,71,83} The final decision could be made, postponed, or be made provisionally on the basis of contextual factors; predicting how parents will feel at a later date and how a context might evolve is impossible, so parents and health-care professionals should maintain communication through follow-up conversations.^{64,83}

Environmental and organisational aspects

Before the decision-making process begins, health-care professionals should prepare themselves, create a safe and quiet place, invite important relatives (or let parents do so), show basic politeness (eg, no pagers or phones, know the name of the child, and introduce oneself), and write or formulate a plan and communicate this to others involved.^{1,49,55,58} Repeated meetings with a common contact person are often needed, and ideally should start as early as possible with conversations about values. Time invested allows for building relationships and constructing values.^{12,13,55,57,58,98} Lastly, involving a multidisciplinary team is useful for ensuring a holistic perspective (eg, social

Panel 2: Communicating values—phrases for health-care professionals

Generally, health-care professionals should tolerate silence, listen, and watch for cues in body language.^{1,56,64,79,83,94} Example phrases are given below each goal.

Addressing emotions

- How are you doing right now?¹
- I am sorry that you are going through this.¹
- Tell me how you are feeling?¹
- We can make this decision together now, but you might want to have a bit more time to think about it, or discuss with others.⁹¹
- This must feel like an impossible situation for you.⁹⁵
- I would like to suggest that we try to buy a little more time... to keep in touch and to give you some time to process what I have told you.⁹⁵
- You were upset by what I was telling you, weren't you? Please can you tell me what was distressing you?¹⁶
- Many parents in your situation feel overwhelmed, angry, or sad. These are normal feelings.⁹⁴
- I realise this must be a very difficult time for you, so we will take it one step at a time.⁹⁴

Inviting and empowering parents to participate in decision making and making clear that more than one option is available

- Is it okay if we talk about your options together?⁶⁹
- Is now a good time to talk?^{1,56}
- We face some decisions here.¹
- For some families... While for other families...^{83,96}
- As parents you are experts on how the condition affects your child.⁸
- Are there other important people that you want to talk to in making this decision?⁹⁷
- There are (for example) two routes that we can take in this situation, we do not know exactly which suits you the most.⁹¹
- There is no question that you know what's best for them.⁹⁵
- Is this a good time?⁹⁴
- Would you want somebody else to be here or for us to call that person?⁹⁴
- Thank you for pointing this out, you know [the child] so well.⁹⁴

Addressing meta values

- Responding to cues such as "this is too hard—I cannot make the decision", or "what would you do, if this was your child?"⁵⁶
- Some parents would like recommendations from me but others prefer to make the decision on their own, where do you fall for this decision?^{83,91}
- Would you like me to share what I see as your most reasonable medical options?⁶⁹

- You do not have to decide.²¹

Open eliciting of values

- What are you hoping for your child, family, or yourself?^{1,56,83}
- What is your biggest fear or worry?^{1,39,56,68,83}
- Do you regret anything that has been done?¹
- What are the main problems for your child or your family?¹
- What gives your life meaning?⁶⁸
- How does your child look to you?⁵⁶
- What are your concerns if we go down path A?⁸³
- What is most important to you (as a family) in what I have just said?^{39,69}
- What is your first feeling or intuition?⁶³
- Do you have all the information you think you need to weigh up these two options?⁹⁷
- If you think about what we have just discussed, what are the most important things for you in this decision?⁹¹
- How do you feel that the benefits of option A relate to the benefits of option B? And what about the disadvantages?⁹¹

Acknowledging uncertainty

- Unfortunately, we cannot be completely sure what will happen to your child.⁹¹
- Many parents I speak to find this uncertainty difficult to deal with.²¹

Reflecting and checking on values

- So, what I am hearing you say is that your child seems to be in pain (after describing the parent's descriptions).¹
- It sounds like factor A is important to you, which means...⁶⁸
- What does it mean to you to be a good parent now that we have this information?⁸⁴
- It sounds like you are saying... (repeat the parent's words)—do I have this right?⁶⁹

Gently challenging preferences

- For some parents a specific percentage of risk seems like a lot, but for others it seems small compared with the benefit.⁶²
- Although you have some negative images of what it would be like to live with a diagnosis, you might be surprised that most patients report a high quality of life.⁶²
- Is this what you most want?⁶⁹
- The real-life experience of disability can be very different than it sometimes seems to people who are not disabled.²⁷
- If we disregard the acute problems and focus on the long-term problems, your child...⁹⁵

Addressing conflicts between participants on values and preferences

- It seems that each of us is focusing on a particular aspect of this situation.⁶⁴

(Continues on next page)

(Panel 2 continued from previous page)

- We can incorporate your standpoint into the discussion that we are having as your child's medical team.⁹⁵
- Of course, I understand that this decision is against your instincts as a parent, but we focus on what would be best in the interests of your child, not in our own interests.¹⁶

Finding commonalities

- We are both hoping for that goal.^{1,56}
- This means that we would not initiate resuscitation; we have tried, and that we really did everything we could.²¹
- I also wish that your child had not been born preterm.⁹⁴

Being supportive (telling parents there are no wrong or right decisions)

- Frame foregoing treatment as equally loving.⁸³
- Loving parents make different decisions about where to go from here.⁸³
- Preventing doing things to your child instead of for them.⁸³
- Parents can have different considerations about what is most valuable to them.⁹¹
- Some parents prefer option A, while others prefer option B. What kind of parent are you?⁹⁶

- The good choice is the one you feel is the best for you and your child.⁹⁴

Translating values towards a choice

- Shall we consider now what we should do as our next step?⁶⁹
- Based on your concern about option A, I think option B at this time might be the next step.⁸³
- Based on what you are telling me I am afraid we are not going to get him to an acceptable quality of life.⁶⁰
- Taking everything into consideration, I think that the best option would be if we decide to...⁹¹
- Discussing follow-up of a decision (eg, decision made, provisional decision, or postpone decision)
- In our next visit, we will...⁸³
- We can always update the plan based on what you are experiencing and how you are feeling about it.⁸³
- We can make the decision together now, but you might want to have a bit more time to think about it and discuss your decision with others. I can come back later for another conversation with you. What is best for you?⁹¹

workers, chaplains, palliative care teams, midwives, nurses, or therapists).^{12,57,58,81}

More broadly, the health-care system should facilitate value clarification and ensure that such clarification is economically viable.⁶⁴ Guidelines could be adapted to encourage this, for example by moving away from universal recommendations and explicitly describing options and trade-offs, or incorporating tools.^{82,99} Digital information, multimedia, and patient decision aids could be helpful in providing information on health-care conditions and potential care options in a relatively unbiased way that avoids problems associated with choice architecture.⁶⁴ These tools should supplement communication—not replace it—and be used by parents who want to use them. One study reported benefits from a patient empowerment campaign that displayed three questions (ie, “what are my options?”, “what are the benefits and harms?”, and “how likely are these?”) in waiting rooms.⁹⁷

Patient decision aids include information about the decision and are specifically designed to help with complex choices, but they could also help with value clarification.^{40,44,47} Two common value clarification methods in patient decision aids are asking people to think about pros and cons and to rate the importance of option attributes. Although these methods reduce decisional conflict, they did not increase value congruence when tested in isolation from conversations.⁴⁷ The most promising methods explicitly and dynamically show people how options align or fail to align with their values.⁴⁷ However, the best approach for a patient decision aid might be contextual.⁴⁰ Explicit value clarification methods

are most useful in well defined decisions, when information is available on common attributes and how they relate to each other and to each option. However, little flexibility in unique and less well described decisions can be a disadvantage of these methods.⁴⁰ Also, attributes can be too general, too uncertain, or not even known and could therefore introduce bias.^{40,65}

Training, education, and understanding

Value clarification skills for health-care professionals might currently be insufficient but can be learned.⁶⁴ Communication training might be particularly useful, as well as education on common factors identified by parents as central to their decision-making process, peer-to-peer coaching, and interviews with families.^{19,49,57,60,64,67} These and other training modalities should be incorporated in medical education programmes and in ongoing training. Practical training and education goals, and a clear normative overview of the terms used in the field of value clarification, would also be helpful.

Conclusions

Incorporating parental values in complex medical decisions for young children is important but challenging. Decisional values are informed by global and external values and could relate to the child, the parents, and the family as a whole. These values should inform preferences and ensure value-congruent choices. Furthermore, parents might hold various meta values on the decision making itself, which should be acknowledged. Complex decisions for young children are emotionally taxing, ethically difficult, and often

Search strategy and selection criteria

A semistructured search was done of PubMed. We aimed to find papers on definitions and explanation of the term value; the concept of value clarification, including its justification and challenges; and guidance for practice. No limitation on dates was used. Search terms were “shared or complex decision-making”, “pediatrics”, “value-eliciting”, and “value-clarification” and synonyms. The search was extended by checking citations and similar articles. After the first screening (April 20, 2022; 90 relevant full-text papers), relevant sections were extracted and summarised thematically. Specific literature was searched to complement and strengthen each part of our manuscript, and a search update was done in January, 2023 (for papers published between April 20, 2022, and Jan 25, 2023; 44 additional relevant full-text papers). Of this total of 134 relevant papers, 94 were ultimately included and cited in this Review. Finally, on the basis of reviewers’ suggestions, five relevant manuscripts were added.

surrounded by uncertainty. These contextual factors make it more likely that values and preferences are initially absent or unstable and will need to be constructed or stabilised. Health-care professionals and parents should work together to construct and clarify values and incorporate them into personalised decisions. An open communication style, with unbiased and tailored information in a supportive environment, is helpful. Dedicated training in communication and shared decision making could improve the incorporation of parental values in complex decisions for young children.

Contributors

RG and DJCW conceptualised and discussed the topic, collected and reviewed relevant literature, and wrote the manuscript. They both approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

Declaration of interests

DJCW is a member of the Royal College of Paediatrics and Child Health Ethics and Law Committee, and of the British Medical Association Medical Ethics Committee. RG declares no competing interests.

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